Can AI Develop a Slicer?

RQ1 · Feasibility

Can an experienced FDM 3D-printer user with no slicer or geometric-computing expertise develop a working slicer using only AI?

RQ2 · Performance

How does the AI-built slicer compare to professional software (Ultimaker Cura) in terms of dimensional accuracy and print time?

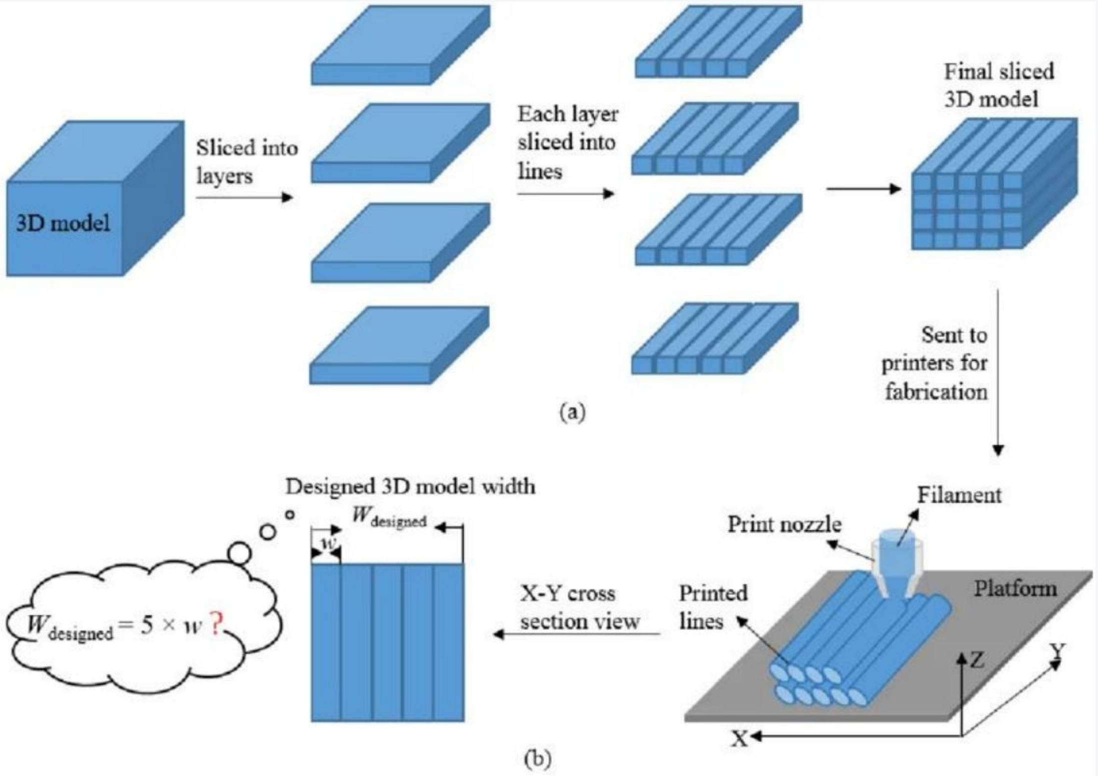
What is a slicer?

A slicer is the compiler. It turns a 3D model into machine instructions (G-code) — the way ***gcc*** turns C into assembly.

Input — the STL file: just a **soup of triangles** with no idea which form a wall or a hole. Rebuilding that structure is the hard part.

Example entry triangle in an STL file:

```
facet normal 0 0 -1
  outer loop
    vertex 0 0 0   /   vertex 10 0 0   /   vertex 10 10 0
  endloop
endfacet
```



Model ⟶ layers ⟶ printer toolpaths. (Jin et al., 2020, CC BY 4.0)

Building the slicer with AI

STL

triangle soup

→

JSON

ordered contours

→

G-code

printer commands

~2,700 lines of Go — every line written by AI from natural-language prompts.

44

prompts kept — out of 150+ issued

30 remain in the live code

AI models used

GPT-5.3 Codex

early development phase

GPT-5.4-mini

later phase

Results: the numbers

Mean deviation from nominal (in mm, rounded) with per-axis breakdown, and mean print time, vs. Ultimaker Cura.

Object · Dev. / Time / per-axis	Cura matched	Vibe Slicer	Cura opt. (ref.)
cube outer	**+0.06 / 09:04** X +0.02 Y +0.13 Z +0.02	**+0.06 / 07:47** X +0.03 Y +0.15 Z +0.01	**−0.01 / 11:32** X −0.03 Y −0.01 Z +0.01
Hole outer	**+0.08 / 28:33** X −0.02 Y +0.30 Z −0.05	**+0.07 / 46:12** X −0.08 Y +0.30 Z −0.01	**+0.03 / 31:47** X −0.03 Y +0.16 Z −0.03
10 mm hole	**−0.05** X −0.07 Y −0.03	**−0.10** X −0.17 Y −0.02	**−0.01** X −0.01 Y 0.00
20 mm hole	**−0.07** X −0.18 Y +0.04	**−0.08** X −0.16 Y +0.01	**−0.04** X −0.09 Y +0.02

Summary Vibe Slicer: matched accuracy under matched parameters; 14% faster on the cube, 62% slower on the holed plate.

Conclusion

RQ1 · Feasibility

Yes* — a working slicer can be built entirely by prompting.

* Very challenging without G-Code experience
* Implementing complex shapes purely AI-based proved very difficult with the AI models used.

RQ2 · Performance

Comparable accuracy on simple geometry; significantly longer print time on complex parts.

Same parameters → matched accuracy and a faster cube; 62% slower print time on the holed plate.

Open question: *would a different AI model — or handing the AI a geometric library (Clipper2) — handle complex shapes better?*

Next — live demo.